1. Basics of Statistics

1. What is Statistics?

Answer:
Statistics is the science of **collecting, organizing, analyzing, interpreting, and presenting data** to make decisions or conclusions.

Example:
Analyzing student marks to understand class performance.

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2. What are the two main types of Statistics?

Answer:

1. **Descriptive Statistics**
2. **Inferential Statistics**

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3. What is Descriptive Statistics?

Answer:
Descriptive statistics summarizes and describes the main features of a dataset.

Examples:

- Mean
- Median
- Mode
- Graphs
- Charts

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4. What is Inferential Statistics?

Answer:
Inferential statistics uses **sample data to make predictions or conclusions about a population**.

Examples:

- Hypothesis testing
- Regression analysis
- Confidence intervals

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2. Population vs Sample

5. What is Population?

Answer:
Population is the **entire group of individuals or observations** that we want to study.

Example:
All students in a university.

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6. What is a Sample?

Answer:
A sample is a **subset of the population** selected for analysis.

Example:
100 students selected from a university.

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7. Why do we use samples instead of populations?

Answer:

Because studying the entire population can be:

- Expensive
- Time consuming
- Sometimes impossible

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8. What is a Sampling Frame?

Answer:
A sampling frame is the **list of all elements in the population from which samples are selected**.

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3. Sampling Techniques

9. What are the main types of sampling techniques?

Answer:

Two categories:

1. Probability Sampling
2. Non-Probability Sampling

Probability Sampling

10. What is Simple Random Sampling?

Answer:
Each member of the population has **equal probability of being selected**.

Example: Lottery method.

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11. What is Systematic Sampling?

Answer:
Selecting every **kth element** from the population.

Example: Every 10th student.

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12. What is Stratified Sampling?

Answer:
Population is divided into **homogeneous groups (strata)** and samples are taken from each group.

Example: Students divided by departments.

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13. What is Cluster Sampling?

Answer:
Population is divided into clusters, and **entire clusters are randomly selected**.

Example: Selecting schools in a city.

Non Probability Sampling

14. What is Convenience Sampling?

Answer:
Samples are selected based on **ease of access**.

Example: Surveying people nearby.

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15. What is Judgment Sampling?

Answer:
Researcher selects samples based on **their knowledge or judgment**.

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4. Variables

16. What is a Variable?

Answer:
A variable is a **characteristic or attribute that can take different values**.

Example:
Age, salary, marks.

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17. What are the two main types of variables?

Answer:

1. Quantitative Variables
2. Qualitative Variables

Quantitative Variables

18. What is a Quantitative Variable?

Answer:
A variable that represents **numerical values**.

Example: Salary, age, height.

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19. Types of Quantitative Variables

1. Discrete
2. Continuous

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20. What is a Discrete Variable?

Answer:
A variable that takes **countable values**.

Examples:

- Number of students
- Number of cars
- Dice outcome

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21. What is a Continuous Variable?

Answer:
A variable that takes **infinite values within a range**.

Examples:

- Height
- Weight
- Temperature
- Time

Qualitative Variables

22. What is a Qualitative Variable?

Answer:
A variable that describes **categories or attributes** rather than numbers.

Example: Gender, city, color.

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23. Types of Qualitative Variables

1. Nominal
2. Ordinal

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24. What is Nominal Data?

Answer:
Data that represents **categories without order**.

Examples:

- Gender
- City
- Blood group

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25. What is Ordinal Data?

Answer:
Data that represents **categories with meaningful order**.

Examples:

- Ratings (1–5 stars)
- Rank in class
- Satisfaction level

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5. Frequency

26. What is Frequency?

Answer:
Frequency is the **number of times a value appears in a dataset**.

Example:

Marks = 60 appears 5 times
Frequency = 5

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27. What is a Frequency Distribution?

Answer:
A frequency distribution is a table that shows **how often each value occurs**.

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28. What is Relative Frequency?

Answer:
Relative frequency is the **proportion of observations belonging to a category**.

Formula:

Relative Frequency = Frequency / Total Observations

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29. What is Cumulative Frequency?

Answer:
Cumulative frequency is the **sum of frequencies up to a certain value**.

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6. Measure of Central Tendency

30. What is Central Tendency?

Answer:
Central tendency refers to the **central or typical value of a dataset**.

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31. Types of Central Tendency

1. Mean

2. Median
3. Mode

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32. What is Mean?

Answer:
Mean is the **average of a dataset**.

Formula:

Mean = Sum of values / Number of values

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33. What is Median?

Answer:
Median is the **middle value when data is arranged in order**.

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34. What is Mode?

Answer:
Mode is the **most frequently occurring value**.

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7. Outliers

35. What is an Outlier?

Answer:
An outlier is a data point that is **very different from other observations**.

Example:
Most salaries = 30k–40k
One salary = 5 lakh → outlier

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8. Five Number Summary

36. What is Five Number Summary?

Answer:
It describes the distribution of data using five values:

1. Minimum
2. Q1 (First Quartile)
3. Median
4. Q3 (Third Quartile)
5. Maximum

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37. What is Interquartile Range (IQR)?

Answer:

IQR = Q3 − Q1

It measures the **spread of the middle 50% of data**.

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9. Measure of Dispersion

38. What is Dispersion?

Answer:
Dispersion describes **how spread out the data values are**.

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39. Types of Dispersion

1. Range
2. Variance
3. Standard Deviation

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40. What is Range?

Answer:

Range = Maximum − Minimum

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41. What is Variance?

Answer:
Variance measures the **average squared deviation from the mean**.

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42. What is Standard Deviation?

Answer:
Standard deviation measures **how much data values deviate from the mean**.

Small SD → data close to mean
Large SD → data spread out

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10. Probability

43. What is Probability?

Answer:
Probability is the measure of **likelihood of an event occurring**.

Formula:

P(E) = Favorable outcomes / Total outcomes

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11. Probability Distribution

44. What is a Probability Distribution?

Answer:
A probability distribution describes **how probabilities are assigned to the values of a random variable**.

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12. Types of Probability Distribution

45. Discrete Distribution

Examples:

- Binomial distribution
- Poisson distribution

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46. Continuous Distribution

Examples:

- Normal distribution
- Log-normal distribution
- Exponential distribution

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13. Normal Distribution

47. What is Normal Distribution?

Answer:
Normal distribution is a **symmetric bell-shaped probability distribution**.

Properties:

Mean = Median = Mode.

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14. PMF, PDF, CDF

48. What is PMF?

Probability Mass Function used for **discrete variables**.

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49. What is PDF?

Probability Density Function used for **continuous variables**.

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50. What is CDF?

Cumulative Distribution Function gives:

$P(X \leq x)$